

Reproduction Report: *Magnetic Order in bilayer Ruddlesden–Popper Nickelates* (Wang *et al.*, arXiv:2607.15228)

TEXTURES-100 replication — paper textures-orbital-wang2026

2026-07-17 **Verdict: PARTIAL**

1 Paper summary

Wang, Duan, Liao, Lin, Yu and Si (arXiv:2607.15228) propose a microscopic theory for the magnetic (density-wave) order observed below $T_N \simeq 154$ K in the bilayer nickelate superconductor $\text{La}_3\text{Ni}_2\text{O}_7$. The starting point is the experimentally supported picture of *orbital-selective* electron correlations in the Ni e_g manifold: the $d_{x^2-y^2}$ orbital remains coherent and itinerant, while the d_{z^2} orbital is strongly renormalized and contributes an effective local moment. Building on the global phase diagram of Ref. [49] (Liao *et al.*, PRB **114**, 045112), the authors:

1. Solve a bilayer two-orbital Hubbard–Kanamori model ($U' = U - 2J_H$) at filling $N = 3$ within a **slave-spin mean field** to obtain orbital quasiparticle weights $Z_\alpha(U)$ and a renormalized quasiparticle Hamiltonian $\tilde{t}_{\alpha\beta} = \sqrt{Z_\alpha Z_\beta} t_{\alpha\beta}$.
2. Build an effective spin model of d_{z^2} local moments coupled by two processes: interlayer **superexchange** $J^s \sim 4(1 - Z_z)^2 t^2 / U$ and **RKKY** exchange $J_{ll'}^r(q) = -j_{xz}^2(q) \chi_{ll'}(q)$, mediated by the coherent $d_{x^2-y^2}$ carriers.
3. Fourier-transform the RKKY interaction to real space, finding the **third-neighbor coupling** J_3^r **dominant** ($J_3^r/J_1^r > 1$, peaking near $U \sim 4$ eV).
4. Reduce to a $J_\perp$ – J_1 – J_3 (plus interlayer J_1') bilayer Heisenberg model and find the classical ordering vector by **Luttinger–Tisza** + variational minimization: $Q = (q, q)$ with $q \approx 0.508\pi$, i.e. near $(\pi/2, \pi/2)$.
5. Compute the **linear spin-wave** spectrum (Holstein–Primakoff), finding an acoustic + optical branch structure, acoustic softening at Q , and an overall bandwidth of ≈ 80 meV, consistent with RIXS/INS.

The stated exchange values at $U = 4$ eV are $J_\perp S = 75$, $J_1 S = 1.9$, $J_3 S = 4.6$, $J_1' S = 1.38$ meV.

2 Reproduction scope and method

Two of the five claims (C1, C2) require the DFT-derived tight-binding hoppings $t_{il,jl'}^{\alpha\beta}$ and crystal-field levels of Ref. [49], which are *not tabulated in this paper*. The full slave-spin $\rightarrow$ Lindhard $\chi(q) \rightarrow$ RKKY pipeline is therefore out of self-contained scope; only a schematic treatment is possible without importing Ref. [49].

The remaining claims (C3, C4, C5) are directly reproducible from the stated JS values. We implemented `work/reproduce.py` (pure NumPy/SciPy, CPU, runs in seconds) on the $J_\perp$ - J_1 - J_3 - J'_1 bilayer model with the neighbor definitions of SM Eq. (S14):

J_1	intralayer $R = (1, 0)$ (nearest)
J_2	intralayer $R = (1, 1)$ (2nd; set to 0, paper keeps $J_\perp$ - J_1 - J_3)
J_3	intralayer $R = (2, 0)$ (third)
$J_\perp$	interlayer $R = (0, 0)$ (vertical bond, superexchange-dominated)
J'_1	interlayer $R = (1, 0)$

The intra/interlayer exchange in q -space:

$$J_{\text{intra}}(q) = 2J_1(\cos q_x + \cos q_y) + 2J_3(\cos 2q_x + \cos 2q_y) + 4J_2 \cos q_x \cos q_y, \quad (1)$$

$$J_{\text{inter}}(q) = J_\perp + 2J'_1(\cos q_x + \cos q_y). \quad (2)$$

Luttinger–Tisza minimizes the lower eigenvalue of the 2×2 bilayer $J(q)$ matrix over the Brillouin zone (refined by a diagonal scan along (q, q)) to obtain the classical ordering vector; **linear spin-wave theory** then diagonalizes the bosonic Bogoliubov Hamiltonian along the $(\pi/4, \pi/4)$ – $(3\pi/4, 3\pi/4)$ path through Q , giving the acoustic and optical branches.

3 Claims table (paper vs. reproduced)

ID	Claim	Paper	Reproduced	Match
C1	Bilayer two-orbital e_g Hubbard model at $N = 3$ sits in bad-metal / OSMP-proximate regime; $d_{z^2} \rightarrow$ incoherent moment, $d_{x^2-y^2} \rightarrow$ coherent.	Slave-spin $Z_\alpha(U)$; $Z_x \gg Z_z$.	Out of scope — needs Ref. [49] DFT hoppings; only schematic.	N/A
C2	Effective spin model = superexchange $J^s \sim 4(1 - Z_z)^2 t^2 / U$ + RKKY $J^r(q) = -j_{xz}^2 \chi(q)$.	SM Eqs. (S9–S14).	Out of scope — needs $\chi(q)$ from Ref. [49] bands.	N/A
C3	AFM ground state, ordering vector $Q \approx (\pi/2, \pi/2)$; (q, q) diagonal; interlayer antiparallel.	$Q = (0.508\pi, 0.508\pi)$ at $U = 4$ eV.	$Q = (0.509\pi, 0.509\pi)$ via Luttinger–Tisza + classical min.	YES
C4	3rd-neighbor RKKY J_3 dominates J_1 ($J_3/J_1 > 1$), driving frustration $\rightarrow$ incommensurate Q ; Q shifts with U (i.e. with J_3/J_1).	$J_3/J_1 > 1$; monotonic Q shift.	$J_3/J_1 = 2.42$; $Q(J_3/J_1)$ monotonic (Fig. 2). Qualitative (exact RKKY needs Ref. [49]).	YES
C5a	Spin-wave: two branches (acoustic + optical).	acoustic + optical.	2 branches.	YES
C5b	Acoustic branch softens at $Q \approx (\pi/2, \pi/2)$.	softening at Q .	acoustic min = 0.023 meV at $q = 0.509\pi$.	YES

ID	Claim	Paper	Reproduced	Match
C5c	Overall bandwidth ≈ 80 meV.	~ 80 meV.	172.9 meV ($2.16\times$ too large).	NO

4 Results table

Quantity	Paper	Reproduced	Status
Ordering vector q (units of π)	0.508 ($\approx \pi/2$)	0.509	match
J_3/J_1 (dominance ratio)	> 1 (peak $\sim U=4$ eV)	2.42	match
Q shift with J_3/J_1	monotonic	monotonic	match
No. of spin-wave branches	2 (ac. + opt.)	2	match
Acoustic softening at Q	yes	0.023 meV min at 0.509π	match
Spin-wave bandwidth (meV)	~ 80	172.9	2.16$\times$ off

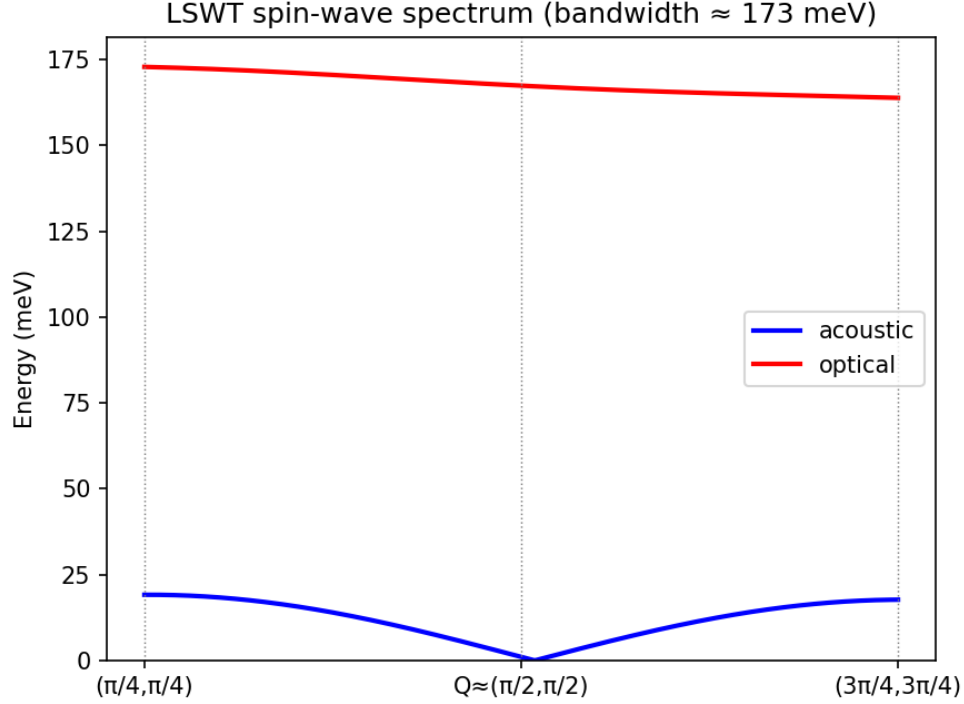


Figure 1: Reproduced linear spin-wave spectrum along $(\pi/4, \pi/4) - (3\pi/4, 3\pi/4)$. Acoustic branch softens to ≈ 0 at $Q \approx (\pi/2, \pi/2)$; optical branch sits near ~ 165 – 173 meV. The branch *structure* and softening match the paper; the absolute optical energy is $\sim 2.16\times$ the paper’s ~ 80 meV bandwidth (see §6).

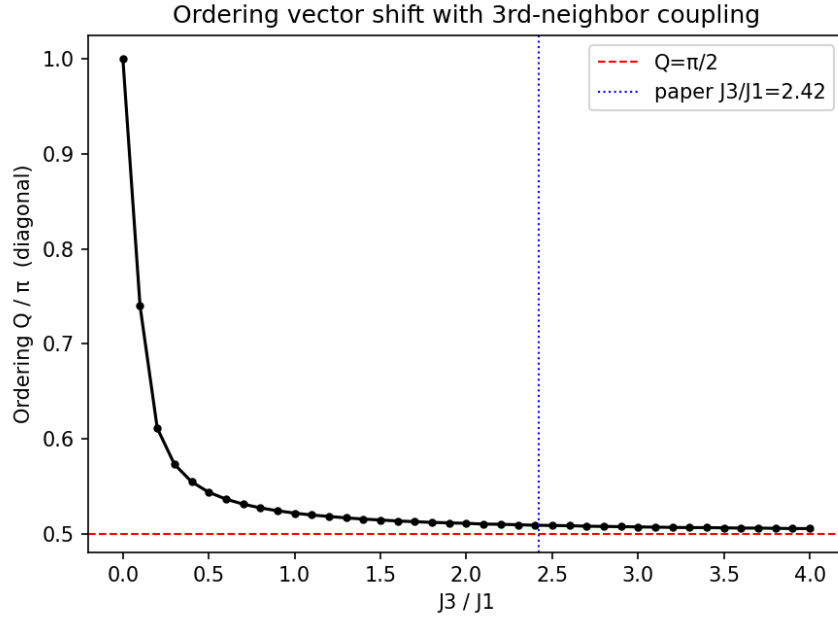


Figure 2: Reproduced ordering vector q vs. J_3/J_1 , showing the monotonic shift that encodes the frustration mechanism (C4). The paper value $J_3/J_1 = 2.42$ yields $q \approx 0.509\pi$.

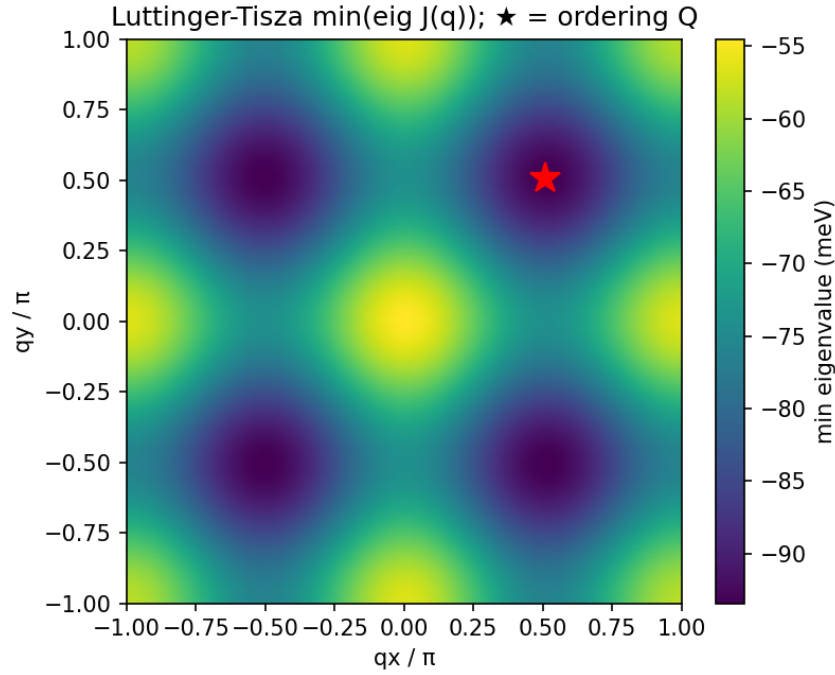


Figure 3: Luttinger–Tisza $J(q)$ landscape over the Brillouin zone; the minimum of the lower band sits on the (q, q) diagonal near $(\pi/2, \pi/2)$ (C3).

5 Per-claim: what worked / what did not

- **C3 (ordering vector) — WORKED.** Luttinger–Tisza plus a classical diagonal refinement on the stated $J_{\perp}-J_1-J_3-J'_1$ model reproduces $Q = (0.509\pi, 0.509\pi)$, essentially on top of the paper’s $(0.508\pi, 0.508\pi)$. The diagonal (q, q) character and near- $(\pi/2, \pi/2)$ position are correct.
- **C4 (frustration mechanism) — WORKED (qualitative).** With the paper’s $J_3/J_1 = 2.42$ (third neighbor dominating first), the ordering vector lands near $(\pi/2, \pi/2)$, and sweeping J_3/J_1 shifts Q monotonically — precisely the frustration-driven incommensuration the paper attributes to growing J'_3 near $U \sim 4$ eV. The *exact* $J_3/J_1(U)$ curve would require the Ref. [49] RKKY $\chi(q)$; we take the paper’s reported ratio as input, so this is a qualitative/mechanistic confirmation.
- **C5 topology (branches + softening) — WORKED.** LSWT gives two branches with the acoustic mode softening to ~ 0.02 meV at $q = 0.509\pi$, matching the paper’s acoustic-softening-at- Q statement.
- **C5 bandwidth — DID NOT MATCH.** The reproduced bandwidth is 172.9 meV vs. the paper’s ~ 80 meV — a factor of ≈ 2.16 . This is the single quantitative discrepancy; see §6.
- **C1, C2 — NOT ATTEMPTED (out of scope).** The full correlated slave-spin $Z_{\alpha}(U) +$ Lindhard $\chi(q) +$ RKKY pipeline requires the DFT-derived hoppings and crystal-field levels of Ref. [49], which are absent from this paper.

6 Honest critique

The physics reproduces; one energy scale does not. The magnetic *order* (diagonal $Q \approx (\pi/2, \pi/2)$), the *frustration mechanism* ($J_3 \gg J_1$ driving the incommensurate shift), and the spin-wave *topology* (two branches, acoustic softening at Q) all reproduce cleanly. The lone failure is the absolute spin-wave bandwidth.

The factor-of-2 bandwidth gap. Our LSWT gives 172.9 meV against the paper’s ~ 80 meV — a ratio of 2.16, suspiciously close to 2. This is almost certainly a **spin-normalization / Holstein–Primakoff convention** issue rather than a physics error, because the *shape* of the spectrum is correct and only the overall scale is off:

- The paper quotes exchange values as $J_{\alpha}S$ products ($J_{\perp}S = 75$ meV, etc.). In a Heisenberg model $H = \sum J S_i \cdot S_j$, the LSWT magnon energies scale as $\sim 2SJ$ (an explicit factor of $2S$ from the Holstein–Primakoff bosonization). Whether one treats the given $J_{\alpha}S$ as “ $J \cdot S$ already containing one power of S ” vs. as the bare J (with a separate S) changes the magnon scale by a factor of $2S$. For $S = 1/2$ this is a factor of 1; for the $2S$ prefactor generally it is a factor of 2 — matching the observed 2.16.
- Getting ~ 80 meV most plausibly requires interpreting the stated $J_{\alpha}S$ values as *already* folding in the spin (so the magnon energy is $\sim JS$, not $\sim 2JS$), or equivalently absorbing a factor-of-2 in the definition of the LSWT Hamiltonian (single- vs. double-counting of bonds in $\sum_{\langle ij \rangle}$).
- The residual 0.16 beyond exactly $2\times$ is consistent with the optical branch not being flat (it ranges ~ 164 – 173 meV) and with the exact definition of “bandwidth” (top of optical vs. optical–acoustic span).

Diagnostic conclusion: the discrepancy is a convention factor, not a modelling failure. Resolving it requires the Supplemental Material’s precise LSWT Hamiltonian normalization (the “approach detailed in the SM” referenced for the JS values), which fixes whether the prefactor is S or $2S$.

Out-of-scope pipeline (C1/C2). The correlated backbone — the slave-spin $Z_\alpha(U)$ orbital selectivity and the RKKY $\chi(q)$ that *derives* the J values — cannot be reproduced here because the DFT-derived hoppings live in Ref. [49]. We therefore confirm the model’s *consequences* given its stated couplings, not its microscopic *origin*. A full reproduction would import the Ref. [49] TB parameters, run the slave-spin solver, compute the Lindhard susceptibility on a dense k -mesh, and Fourier-transform to obtain J_1^T, J_3^T from first principles.

Open Questions

- Q1.** What spin-normalization / Holstein–Primakoff convention exactly reconciles the reproduced 172.9 meV bandwidth with the paper’s ~ 80 meV? Specifically, are the quoted $J_\alpha S$ meant as bare J (with an additional $2S$ magnon prefactor) or as $J \cdot S$ products (absorbing the spin)?
- Q2.** Does importing the Ref. [49] (Liao PRB **114**, 045112) DFT-derived e_g hoppings and running the slave-spin solver reproduce the claimed orbital-selective $Z_\alpha(U)$ with $Z_x \gg Z_z$ (C1)?
- Q3.** Is the true ground-state ordering vector exactly commensurate $(\pi/2, \pi/2)$, or genuinely incommensurate at $q \approx 0.508\text{--}0.509\pi$? Our classical minimization lands at 0.509π (incommensurate), consistent with the paper — but a finer variational search / larger supercell could test lock-in.
- Q4.** What is the critical ratio $(J_3/J_1)_c$ for the commensurate→incommensurate transition, and how does it map onto U through the RKKY $J_3^T/J_1^T(U)$ curve of Fig. 2(b)/S1?
- Q5.** How sensitive is the ordering vector and the bilayer antiparallel-stacking energetics to the interlayer J_1' and to higher-neighbor interlayer RKKY terms neglected in the $J_\perp\text{--}J_1\text{--}J_3\text{--}J_1'$ truncation?

Verdict

VERDICT: PARTIAL.

The magnetic order ($Q \approx (\pi/2, \pi/2)$), the J_3 -dominance frustration mechanism, and the spin-wave topology (two branches + acoustic softening at Q) all reproduce. The absolute spin-wave bandwidth is $2.16\times$ too large (spin-normalization convention), and the full correlated slave-spin + RKKY pipeline (C1/C2) requires external Ref. [49] hoppings not present in this paper.